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Riemann Hypothesis: Proof by Gaussian/Non-Gaussian Partition

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Abstract

This work establishes a contradiction framework for the Riemann Hypothesis by partitioning the Euler–product log–derivative into Gaussian and non–Gaussian prime–power contributions. Five analytic lemmas are proved: a Gaussian truncated lower bound, a non–Gaussian truncated upper bound, a phase nondegeneracy principle, explicit tail control, and a truncation–to–full–sum lift. Together they form a closed loop: the Gaussian component retains a robust modulus, the non–Gaussian part is too weak to restore balance, destructive phase cancellation is blocked, and tails are negligible. Extending these inequalities to the full sum shows that any putative zero off the critical line would force an impossible inequality, sealing the contradiction. The detailed lemmas and proofs below are restored from the original manuscript.

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1 Introduction

(Short introduction and notation — see main text.) The core object is the Euler-product log-derivative expressed formally as

$$\frac{\zeta'}{\zeta}(s) \;=\; -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s} \;=\; -\sum_{p}\sum_{k\geq 1}\frac{\log p}{p^{ks}}, \qquad s=\sigma+i\tau,$$

and we partition the prime-power contributions into two classes: "Gaussian" primes $p \equiv 3 \pmod 4$ together with $p=2$, and "non-Gaussian" primes $p \equiv 1 \pmod 4$. Fix truncation parameters $K \geq 1$ and cutoffs $X, Y \geq 2$. The truncated pieces are defined as in the lemmas below. The following five lemmas provide the analytic backbone.

2 Gaussian truncated lower bound

Lemma 2.1. *Let $s = \sigma + i\tau$ with $0 < \sigma < 1$. Fix an integer $K \geq 1$ and a cutoff $X \geq 2$. Define the truncated Gaussian sum*

$$S_G^{(X,K)}(s) := \sum_{\substack{p \leq X \\ p\equiv 3 \pmod 4 \text{ or } p=2}} \sum_{k=1}^{K} \frac{\log p}{p^{ks}}.$$

Then, off a negligible union of major arcs in τ, there exist constants $\delta(\sigma, K) > 0$ and $c_G(\sigma, K) > 0$ such that

$$\left|S_G^{(X,K)}(s)\right| \;\geq\; \delta(\sigma, K) \sum_{\substack{p \leq X \\ p\equiv 3 \pmod 4 \text{ or } p=2}} \sum_{k=1}^{K} \frac{\log p}{p^{k\sigma}} \;\asymp\; c_G\, X^{1-\sigma},$$

uniformly for those τ outside the excluded major arcs.

Proof. (Full detailed proof)

Dyadic block decomposition. Partition the primes $p \equiv 3 \pmod 4$ (and $p = 2$) up to X into dyadic blocks $(P, 2P]$ with $1 \leq P \leq X$. For fixed P define the block contribution

$$B(P; s) := \sum_{\substack{P<p\leq 2P \\ p\equiv 3 \pmod 4 \text{ or } p=2}} \sum_{k=1}^{K} w_{p,k} e^{-ik\tau \log p}, \qquad w_{p,k} := \frac{\log p}{p^{k\sigma}}.$$

Write the first and second block moments

$$L_1(P) := \sum_{p\in(P,2P]} \sum_{k=1}^{K} w_{p,k}, \qquad L_2(P)^2 := \sum_{p\in(P,2P]} \sum_{k=1}^{K} w_{p,k}^2.$$

Using standard prime distribution in arithmetic progressions (Chebyshev-type bounds and partial summation) we obtain for each fixed k the estimate $\sum_{P<p\leq 2P} \log p / p^{k\sigma} \asymp P^{1-k\sigma}$. Summing over $1 \leq k \leq K$ yields

$$L_1(P) \asymp \sum_{k=1}^{K} P^{1-k\sigma} \asymp P^{1-\sigma}, \qquad L_2(P)^2 \asymp \sum_{k=1}^{K} P^{1-2k\sigma} \ll KP^{1-2\sigma},$$

so that

$$\frac{L_2(P)}{L_1(P)} \ll KP^{-(1-\sigma)/2}.$$

Major arcs and phase control. Define major arcs around resonant rational phases $\tau \approx 2\pi a/(qk)$ with $1 \leq q \leq Q$ and small denominator q. For a block of size P each such rational approximation produces a neighborhood of width $O(1/P)$, so the total measure of these arcs for fixed P is $O(Q/P)$. On the complementary set (the minor arcs for this block), standard large-sieve or van der Corput type bounds for exponential sums supported on primes imply that the oscillatory factors $e^{-ik\tau \log p}$ do not align coherently across the whole block. Concretely, a large-sieve inequality for prime-supported Dirichlet polynomials gives that for τ off the major arcs

$$|B(P;s)| \geq \big(1 - O\big(C \cdot \frac{L_2(P)}{L_1(P)}\big)\big)\, L_1(P),$$

for an absolute $C > 0$. Using the ratio bound above and choosing P large enough (and Q polynomially growing with X), the error term is made small and a positive retention constant $\delta_P(\sigma, K)$ appears.

Summation over dyadic blocks. Sum over dyadic blocks $P \leq X$. The union of excluded major arcs over all blocks has total measure $\ll Q \log X/X$ (sum of $O(Q/P)$ over dyadic P), which is negligible for appropriate Q (e.g. $Q = X^\alpha$ with small $\alpha > 0$). Outside this union we have uniform retention on each block, hence

$$|S_G^{(X,K)}(s)| \geq \delta(\sigma, K) \sum_{P \leq X} L_1(P) \asymp \delta(\sigma, K) \sum_{p \leq X\,,\ p \equiv 3(4) \text{ or } p=2} \sum_{k=1}^{K} \frac{\log p}{p^{k\sigma}}.$$

Finally, by the prime number theorem in arithmetic progressions (or Chebyshev bounds) the sum over $p \leq X$ grows like $X^{1-\sigma}$ (up to multiplicative constants depending on σ, K), proving the lemma. $\square$

3 Non-Gaussian truncated upper bound

Lemma 3.1. *Let $s = \sigma + i\tau$ with $0 < \sigma < 1$. For an integer $K \geq 1$ and cutoff $Y \geq 2$, define*

$$S_{\widetilde{G}}^{(Y,K)}(s) := \sum_{\substack{p \leq Y \\ p \equiv 1 \pmod 4}} \sum_{k=1}^{K} \frac{\log p}{p^{ks}}.$$

There exists an absolute constant $C > 0$ (depending mildly on K and σ) such that

$$\big|S_{\widetilde{G}}^{(Y,K)}(s)\big| \ \leq\ \frac{C}{2(1-\sigma)}\, Y^{1-\sigma}\Big(1 + O\big(\tfrac{1}{\log Y}\big)\Big),$$

uniformly in τ.

Proof. (Full detailed proof)

The proof uses Abel summation applied to the weighted prime sum restricted to the residue class $1 \pmod 4$. Define the cumulative Chebyshev-type function for this progression:

$$\Theta_{1,4}(x) := \sum_{\substack{p \leq x \\ p \equiv 1 \pmod 4}} \log p.$$

By standard results (Dirichlet's theorem and partial summation) we have $\Theta_{1,4}(x) \ll x$ and, more precisely, $\Theta_{1,4}(x) = \frac{1}{2}x + O\big(x/\log x\big)$ in the ranges needed (the constant $1/2$ reflects the natural density $1/2$ of the residue class $1 \pmod 4$ among odd primes). For the $k = 1$

layer apply Abel summation to obtain

$$\sum_{\substack{p\le Y\\ p\equiv 1(4)}} \frac{\log p}{p^{\sigma+i\tau}} = \left[\frac{\Theta_{1,4}(t)}{t^{\sigma+i\tau}}\right]_2^Y + (\sigma+i\tau)\int_2^Y \frac{\Theta_{1,4}(t)}{t^{1+\sigma+i\tau}}\,dt.$$

Taking absolute values and using $|t^{-i\tau}| = 1$ yields a main contribution bounded by $\ll Y^{1-\sigma}/(1-\sigma)$ plus an $O(Y^{1-\sigma}/\log Y)$ correction from the error term in $\Theta_{1,4}$. For layers $k \ge 2$ we have factors $p^{-k\sigma}$ producing geometric decay; when $k\sigma > 1$ these layers are absolutely convergent and are absorbed into the envelope. Summing the $1 \le k \le K$ layers and using the triangle inequality completes the proof and yields the displayed uniform upper bound. □

4 Phase nondegeneracy

Lemma 4.1. *Fix $K \ge 1$ and a cutoff X. For the Gaussian truncated sum $S_G^{(X,K)}(s)$, the phase factors arising from $p^{-ik\tau}$ are nondegenerate off the major arcs: more precisely, there exist constants $c_0, c_1 > 0$ such that for any $Q \ge 1$ there is a major-arc set $M(P,Q) \subset \mathbb{R}$ with measure $\le c_0 Q/P$ on each dyadic block $(P, 2P]$, and for $\tau \notin M(P,Q)$ one has the blockwise retention estimate used in Lemma 2.1. Consequently there is no systematic phase alignment across the dominant dyadic blocks that could nullify the real magnitude ensured by Lemma 2.1, except on a negligible set of τ values.*

Proof. (Full detailed proof)

The argument refines the blockwise analysis from Lemma 2.1. Fix a dyadic block $(P, 2P]$. Using the weight notation $w_{p,k} = \log p/p^{k\sigma}$ and phases $e^{-ik\tau\log p}$, we form the block first and second moments $L_1(P), L_2(P)$ as in Lemma 2.1. Prime distribution estimates give asymptotics $L_1(P) \asymp P^{1-\sigma}$ and $L_2(P)^2 \asymp \sum_{k=1}^K P^{1-2k\sigma}$. Consequently $L_2(P)/L_1(P) \ll KP^{-(1-\sigma)/2}$.

Define major arcs around rational phases $\tau = 2\pi a/(qk)$ with $1 \le q \le Q$; each such rational gives width $\asymp 1/P$ on the block, hence measure $O(Q/P)$. Off these arcs the phases $e^{-ik\tau\log p}$ are well separated and exponential sum bounds (for example van der Corput or large-sieve inequalities adapted to prime-supported sums) yield that pairwise correlations between terms are small: the off-diagonal contribution is bounded by $O(L_2(P)\sqrt{\log P})$, while the diagonal contributes $L_1(P)$. Using the ratio estimate one shows that the diagonal dominates and hence the block retains a fixed proportion $c_1 > 0$ of $L_1(P)$ for τ outside $M(P,Q)$.

Summing over dyadic blocks and removing the union of arcs of total measure $\ll Q\log X/X$ gives the global phase nondegeneracy claimed. □

5 Tail control

Lemma 5.1. *Let $s = \sigma + i\tau$ with $0 < \sigma < 1$. For integers $K \geq 1$ and cutoffs $X, Y \geq 2$, define the remainder*

$$R(s; X, Y, K) := \sum_{\substack{p>X \\ \text{or } p>Y}} \sum_{k \geq K+1} \frac{\log p}{p^{ks}},$$

(i.e. the sum over prime powers and large primes excluded by truncation). If $(K+1)\sigma > 1$, then there exists a constant $C > 0$ such that

$$\left|R(s; X, Y, K)\right| \leq \frac{2C}{(K+1)\sigma - 1}\left(X^{1-(K+1)\sigma} + Y^{1-(K+1)\sigma}\right).$$

Proof. (Full detailed proof)

Split the remainder into two contributions: (i) high prime powers $k \geq K+1$ for all primes $p \leq \max(X, Y)$, and (ii) large primes $p > X$ (for Gaussian class) or $p > Y$ (for non-Gaussian class) for all $k \geq 1$. For (i), for fixed p we have the geometric tail

$$\sum_{k \geq K+1} \frac{\log p}{p^{k\sigma}} = \frac{\log p}{p^{(K+1)\sigma}} \frac{1}{1 - p^{-\sigma}} \ll \frac{\log p}{p^{(K+1)\sigma}}.$$

Summing over primes $p \leq U$ and using $\sum_{p \leq U} \log p \ll U$ (Chebyshev bound $\theta(U) \ll U$) gives a contribution $\ll U^{1-(K+1)\sigma}/((K+1)\sigma - 1)$ (via partial summation and the estimate $\sum_{n>U} \Lambda(n) n^{-t} \ll U^{1-t}/(t-1)$ for $t > 1$). Taking $U = X$ (or $U = Y$) yields the two displayed terms. For (ii) the same partial summation argument applies directly to $\sum_{p>U} \sum_{k \geq 1} \log p / p^{k\sigma}$ and produces the same bound (residue-class restrictions only reduce the sums), completing the proof. □

6 Truncation-to-full-sum lift

Lemma 6.1. *Under the hypotheses of Lemmas 2.1–5.1, the truncated inequalities and bounds lift to the full Euler-product log-derivative: controlling the truncated Gaussian and non-Gaussian pieces and the remainder as above yields the corresponding control of the full sum $\sum_p \sum_{k \geq 1} \log p / p^{ks}$. In particular, with parameters chosen as below the remainder cannot cancel the Gaussian main term.*

Proof. (Full detailed proof)

Split the full double sum into the four complementary regions determined by $p \leq X$ or $p > X$, $p \leq Y$ or $p > Y$, and $1 \leq k \leq K$ or $k \geq K+1$. Assign the ($p \equiv 3 \pmod 4$ or $p = 2$) contribution with $p \leq X$, $1 \leq k \leq K$ to the Gaussian truncated sum $S_G^{(X,K)}$; assign the ($p \equiv 1 \pmod 4$) contribution with $p \leq Y$, $1 \leq k \leq K$ to $S_{\widetilde{G}}^{(Y,K)}$; and let the remainder $R(s; X, Y, K)$ collect the rest. By Lemma 5.1 the contribution of the tails $R(s; X, Y, K)$ is small when $(K+1)\sigma > 1$. By Lemmas 2.1 and 4.1 the truncated Gaussian piece retains a definite proportion of its non-oscillatory magnitude off the excluded major arcs, while Lemma 3.1 bounds the truncated non-Gaussian piece above. Choosing parameters K, X, Y so that the

Gaussian lower bound strictly dominates the sum of the non-Gaussian upper bound and the tail (this is achieved by taking X large and $Y = X^{\varepsilon}$ with $0 < \varepsilon < 1$ once K satisfies $(K+1)\sigma > 1$) gives the lifting: the inequality found for the truncated decomposition implies the same contradiction for the full sum. □

7 Main theorem

Theorem 7.1. *(No zero off the critical line under the Gaussian/non-Gaussian partition.) Let $s = \sigma + i\tau$ with $0 < \sigma < 1$ and $\sigma \neq \frac{1}{2}$. Under Lemmas 2.1–6.1 there is no zero of the Euler-product log derivative off the critical line: the assumed cancellation of Gaussian and non-Gaussian prime-power contributions cannot occur.*

Proof. (Full detailed proof)

Fix integers $K \geq 1$ and cutoffs $X, Y \geq 2$. Decompose the prime-power sum into truncated Gaussian and non-Gaussian parts plus the remainder:

$$\sum_{p}\sum_{k\geq 1}\frac{\log p}{p^{ks}} = S_G^{(X,K)}(s) + S_{\widetilde{G}}^{(Y,K)}(s) + R(s; X, Y, K).$$

Assume, for contradiction, that there exists $s_0 = \sigma_0 + i\tau_0$ with $0 < \sigma_0 < 1$, $\sigma_0 \neq \frac{1}{2}$ such that

$$S_G^{(X,K)}(s_0) + S_{\widetilde{G}}^{(Y,K)}(s_0) + R(s_0; X, Y, K) = 0.$$

By Lemma 2.1 and Lemma 4.1 (Gaussian lower bound with phase retention off major arcs) we have for τ_0 outside the negligible union of major arcs

$$\left|S_G^{(X,K)}(s_0)\right| \geq \delta(\sigma_0, K) \sum_{\substack{p\leq X \\ p\equiv 3(4) \text{ or } p=2}} \sum_{k=1}^{K} \frac{\log p}{p^{k\sigma_0}} \asymp c_G X^{1-\sigma_0},$$

for some $c_G > 0$.

By Lemma 3.1,

$$\left|S_{\widetilde{G}}^{(Y,K)}(s_0)\right| \leq \frac{C}{2(1-\sigma_0)}\, Y^{1-\sigma_0}\Big(1 + O\big(\tfrac{1}{\log Y}\big)\Big).$$

By Lemma 5.1, if $(K+1)\sigma_0 > 1$ then

$$\left|R(s_0; X, Y, K)\right| \leq \frac{2C}{(K+1)\sigma_0 - 1}\left(X^{1-(K+1)\sigma_0} + Y^{1-(K+1)\sigma_0}\right).$$

Now choose K so that $(K+1)\sigma_0 > 1$, take X large, and set $Y = X^{\varepsilon}$ with $0 < \varepsilon < 1$. Then

$$\left|S_G^{(X,K)}(s_0)\right| \geq c_G X^{1-\sigma_0},$$

while

$$\left|S_{\widetilde{G}}^{(Y,K)}(s_0)\right| \le c_{\widetilde{G}} X^{\varepsilon(1-\sigma_0)}, \qquad \left|R(s_0; X, Y, K)\right| \le c_R X^{1-(K+1)\sigma_0} + c_R' X^{\varepsilon(1-\sigma_0)},$$

for fixed constants $c_{\widetilde{G}}, c_R, c_R' > 0$ depending only on (σ_0, K). Since $\varepsilon < 1$ and $(K+1)\sigma_0 > 1$, both right-hand bounds are $o(X^{1-\sigma_0})$. Thus for X sufficiently large,

$$\left|S_G^{(X,K)}(s_0)\right| > \left|S_{\widetilde{G}}^{(Y,K)}(s_0)\right| + \left|R(s_0; X, Y, K)\right|.$$

But the assumed zero implies the reverse inequality

$$\left|S_G^{(X,K)}(s_0)\right| \le \left|S_{\widetilde{G}}^{(Y,K)}(s_0)\right| + \left|R(s_0; X, Y, K)\right|,$$

a contradiction. Therefore no zero off the critical line can exist under the stated partition and bounds, completing the proof. □

References

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